CAREFLOW

Clinical Pathway Process Mining

Healthcare Operations & Process Analytics

Project Presentation

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Exploratory analysis using a synthetic patient-event dataset

1. Problem & Motivation

Why look beyond a single average duration?

A single average pathway duration can hide different activity sequences.

Repeated activities and uncommon transitions may be difficult to spot in summary KPIs.

Activity-level elapsed intervals can help identify patterns worth investigating.

CareFlow uses process mining and Power BI to make these patterns easier to explore.

500
Synthetic patient visits

3,744
Recorded events

2. Project Objectives

From event data to an interactive process view

Generate and prepare a timestamped synthetic event log.

Discover and summarize activity sequences and transition frequencies.

Measure pathway duration and identify repeated activities and long pathways.

Check cases against a defined set of pathway sequence rules.

Export analysis outputs to Excel and communicate findings in Power BI.

3. Dataset & Technology Workflow

Synthetic event log prepared for exploratory process mining

500	3,744	7.49	7
Patient visits	Recorded events	Average events / visit	Workbook sheets

Google Colab + Python: event-log preparation and analysis

PM4Py: process-mining analysis and process-flow exploration

Excel: seven-sheet analysis workbook

Power BI Desktop: KPI cards, transition charts, elapsed-time comparisons and slicers

4. Methodology

How the analysis was performed

1 Prepare event log

Case ID, activity and timestamp

2 Explore process

Activity sequence and process flow

3 Measure transitions

Directly-following activity counts

4 Review time patterns

Average, median and maximum elapsed intervals

5 Validate configured rules

Four rule-based sequence checks

6 Present results

Excel workbook and Power BI dashboard

5. Key Performance Indicators

Headline results from the supplied analysis workbook

131.61 min
Average pathway duration

131 min
Median pathway duration

216 min
Maximum pathway duration

38.80%
Visits with repeat Triage

100%
Rule-based conformance

23
Long pathways flagged

All values describe the synthetic dataset and the project's configured analysis rules.

6. Activity Transition Findings

Most frequent directly-following activity pairs

From activity	To activity	Count
Triage	Doctor Consultation	694
Registration	Triage	500
Doctor Consultation	Treatment	500
Treatment	Discharge	500
Doctor Consultation	X-Ray	335
X-Ray	Laboratory Test	147
Laboratory Test	Doctor Consultation	134
X-Ray	Doctor Consultation	115
Laboratory Test	Triage	81
X-Ray	Triage	73

Triage → Doctor Consultation is the most frequent transition (694).

Counts can exceed 500 visits because some simulated pathways repeat activities.

Transition frequency describes how often a pair occurs, not how long it takes.

7. Elapsed-Time Patterns

Highest average intervals in the Time Bottlenecks sheet

Transition	Count	Avg (min)	Median	Max
Doctor Consultation → Triage	40	21.88	22	35
X-Ray → Triage	73	21.67	24	35
Doctor Consultation → X-Ray	335	21.41	23	35
Doctor Consultation → Doctor Consultation	57	20.89	20	35
Treatment → Discharge	500	20.58	21	35

Doctor Consultation → Triage has the highest average elapsed interval: 21.88 minutes.

X-Ray → Triage follows at 21.67 minutes.

These are elapsed intervals between recorded events—not verified queue waiting times.

8. Repetition & Rule-Based Conformance

What the configured checks show

38.80%
Visits with repeat Triage

100%
Cases passing four rules

23
Long pathways flagged

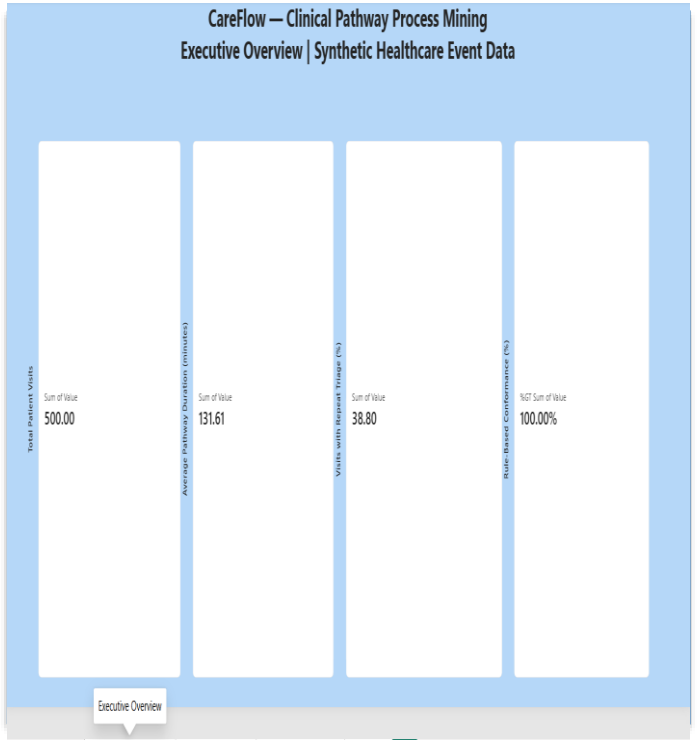
Repeated activities are useful signals for further case-level investigation.

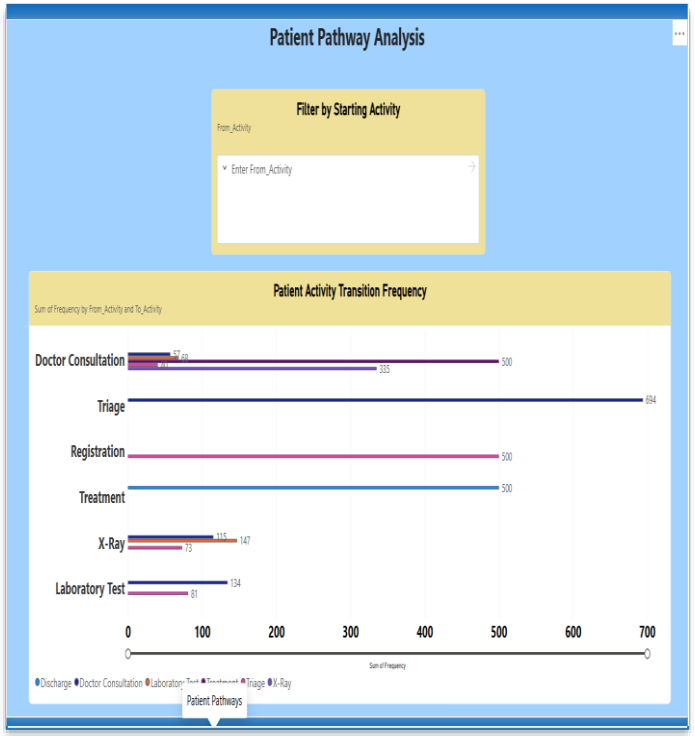
100% conformance means all cases passed the four configured sequence checks only.

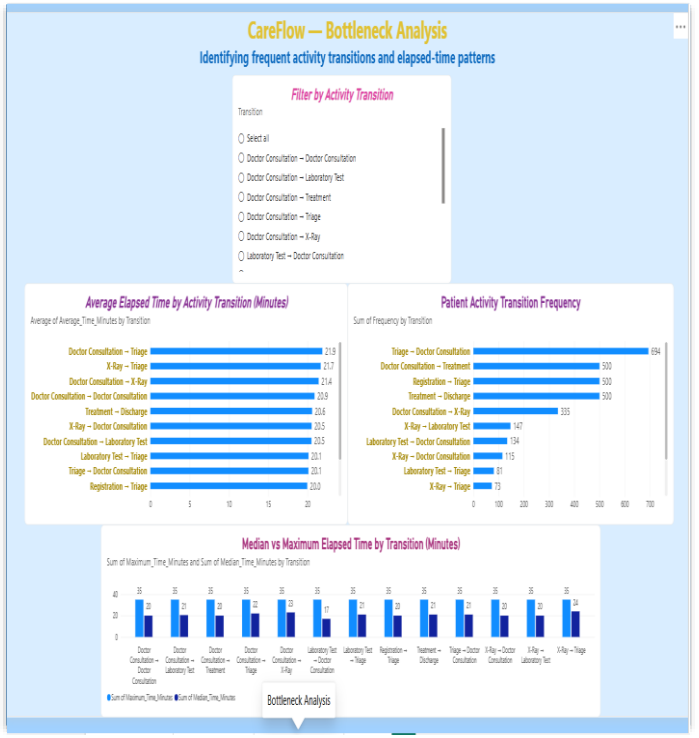
It does not prove clinical quality, safety, or compliance with a complete hospital protocol.

9. Power BI Dashboard

Insert your actual exported screenshots in the three placeholders







The report contains KPI cards, activity-transition analysis, elapsed-time comparisons and interactive slicers.

10. Limitations & Interpretation

Important boundaries on what can be concluded

The event log is synthetic; results are illustrative and cannot be generalized to a real hospital.

Elapsed time between activities is not automatically queue waiting time or service time.

The four conformance rules are simplified sequence checks, not a full clinical protocol.

The analysis does not establish causes for repeated activities or longer pathways.

The configured long-pathway threshold should be documented from the source notebook before submission.

11. Conclusion

CareFlow demonstrates an exploratory process-mining workflow: transforming a synthetic event log into pathway, transition, elapsed-time, repetition and conformance insights, then communicating selected results through an interactive Power BI dashboard.

Next step: validate the workflow and rules with real, governed operational data before drawing conclusions about actual hospital performance.

Thank you